force-directed layout
  adding nodes 197–200
  charge property 193
  creating diagram 186–188
  gravity property 193
  linkDistance property 194
  linkStrength property 194–195
  manually positioning nodes 201–202
  network measures
    centrality 191
    clustering 193
    degree 192–193
    edge weight 190
  optimization 202
  overview 185–186
  removing nodes 197–200
  SVG markers 188–190
  updating network 195–196
formatting data
  categorical 51
  geometric 51
  overview 50
  quantitative 50–51
  raw 52
  temporal 51–52
  topological 51
functions
  array 33–34
  inline 59–61

G

<g> element 23–24
GDAL 207
generators 109
geodata
  creating random 306–309
  drawing with canvas 309–310
  mixed-mode rendering techniques 310–316
  overview 304–306
geographic data 36–37
geographic information systems. *See* GIS
GeoJSON 37, 207–209, 227
geometric data 51
geometric primitives 22
geospatial information
  maps
    canvas drawing 237
    cartograms 238–239
    drawing points on 212–213
    finding data 206–207
    GeoJSON 207–209
    globes 221–226
    graticule 217
    hexbins 238
    interactivity 215–216
    overview 206
    projection 209–210, 213–215
    raster reprojection 238
    satellite projection 226–227
    scale 210–211
    tile mapping 233–235
    transform zoom 237
    Voronoi diagrams 238
    zoom 217–221
  TopoJSON
    file format 227
    merging 229–231
    neighbors 232–233
    overview 227
    rendering 228
Gephi 36, 178
GIS (geographic information systems) 37
globes 221–226
Google Analytics 163
Google Chrome 16
graphical primitives 22
graphical zoom 219
graticule 217
gravity property 193

H

“hairballs” 190
Hello World examples
  with circles 41–42
  conversation 42
  with divs 40–41
hexbins 238
histograms 140–142
hover events 270–274
:hover state 27
hsl() function 90
html() function 33, 48, 97
<html> element 13
HTML5 (Hypertext Markup Language 5)
  CSS 26–29
  DOM
    in console 14–16
    overview 12–14
  JavaScript
    array functions 33–34
    method chaining 30–33
    overview 29–30
  as pregenerated content 95–97
  SVG
    <circle> element 21–23
    <g> element 23–24
    <line> element 21–23
    overview 18–20
    <path> element 25–26
    <polygon> element 21–23
    <rect> element 21–23
    <svg> element 20–21
    <text> element 23
  testing in console 18

I

image gallery example
  interactively highlighting elements 254–255
  overview 252–254
  selecting image 255
images
  as pregenerated content 94–95
  project architecture and 79
indexOf() function 197
information visualization
  binding data
    append() function 59
    attr() function 59
    data() function 58–59
    enter() function 59
    inline functions 59–61
    integrating scales 61–65
    selectall() function 58
    selections and 57–58
  data flow 47
  enter, update, and exit 70–76
  formatting data
    categorical 51
    geometric 51
    overview 50
    quantitative 50–51
    raw 52
    temporal 51–52
    topological 51
  loading data
    formats and 48–50
    overview 47–48
  measuring data 56–57
  setting channels 67–69